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DSA0105-Object Oriented Programming with C++

CO3-AT1 Analytical Problem Solving

Question-1:

Define a class to represent a bank account. Include the following members:

Data members:

1. Name of the depositor

2. Account number

3. Type of account

4. Balance amount in the account

Member functions:

1. To assign initial values

2. To deposit an amount checking with account number and name of the customer

3. To withdraw an amount after checking the balance

4. To display name and balance

Write a main program to test the program use constructors and destructors.

PROGRAM

```
#include<iostream>

using namespace std;


class bank
```

```
{
    string name;
    int acc;
    float bal;

public:
    bank(string n, int a, float b)
    {
        name = n;
        acc = a;
        bal = b;
    }

    void deposit(float x)
    {
        bal += x;
    }

    void withdraw(float x)
    {
        if(x <= bal)
            bal -= x;
    }

    void display()
```

```
    {
        cout << name << endl << bal << endl;
    }

    ~bank()
    {
        cout << "Account closed";
    }
};

int main()
{
    bank a("sri", 101, 5000);

    a.deposit(2000);
    a.withdraw(1000);
    a.display();

    return 0;
}
```

OUTPUT

```
Output

sri
6000
Account closed
```

Question-2:

Use Classes, Object, Constructor and complete the following application.

A book shop maintains the inventory of books that re being sold at the shop. The list includes details such as author, title, price, publisher and stock position. Whenever a customer wants a book, the sales person inputs the title and author and the system searches the list and displays whether it is available or not. If it is not, an appropriate message is displayed. If it is, then the system displays the book details and requests for the number of copies required. If the requested copies are available, the total cost of the requested copies is displayed; otherwise, the message "Required copies not in stock" is displayed. Design a system using a class called books with suitable member functions and constructor.

Also improve the system to incorporate the following features:

(i) The price of the books should be updated as and when required, Use a private member function to implement this.

(ii) The stock value of each book should be automatically updated as soon as a transaction is completed.

(iii) The number of successful and unsuccessful transactions should be recorded for the purpose of statistical analysis.

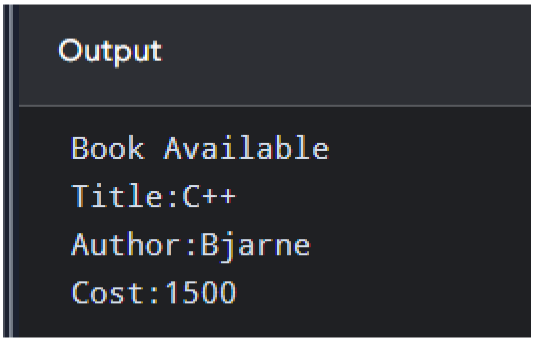
PROGRAM

```
#include<iostream>

using namespace std;


class book
```

```
{
    string title,author;
    float price;
    int stock;

public:
    book(string t,string a,float p,int s)
    {
        title=t;
        author=a;
        price=p;
        stock=s;
    }

    void sell(int n)
    {
        if(n<=stock)
        {
            cout<<"Book Available\n";
            cout<<"Title:"<<title<<endl;
            cout<<"Author:"<<author<<endl;
            cout<<"Cost:"<<price*n<<endl;
            stock-=n;
        }
        else
            cout<<"Required copies not in stock";
    }
};
```

```
int main()

{

    book b("C++","Bjarne",500,10);

    b.sell(3);


    return 0;

}
```

OUTPUT



Question-3:

Create a class FLOAT that contains one float data member. Overload all four arithmetic operators (+,-,*,/) so that they operate on the objects of class FLOAT.

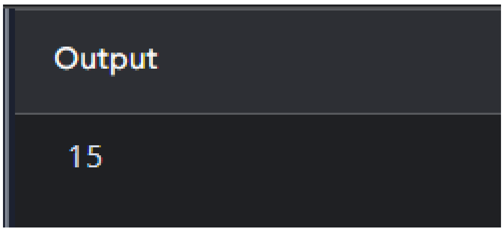
PROGRAM

```
#include <iostream>

using namespace std;


class FLOAT

{

    float x;


public:

    FLOAT(float a)

    {
```

```
        x = a;
    }

    FLOAT operator+(FLOAT b)
    {
        return FLOAT(x + b.x);
    }

    FLOAT operator-(FLOAT b)
    {
        return FLOAT(x - b.x);
    }

    FLOAT operator/(FLOAT b)
    {
        return FLOAT(x / b.x);
    }

    void show()
    {
        cout << x;
    }
};

int main()
{
    FLOAT a(10), b(5), c(0);

    c = a + b;
```

```
    c.show();

    return 0;
}
```

OUTPUT



Question-4:

Define a class String. Use overloaded ==, >, < operators to compare two strings.

PROGRAM

```
#include <iostream>
#include <string>
using namespace std;
class String
{
    string s;

public:
    String(string x)
    {
        s = x;
    }

    bool operator==(String b)
    {
        return s == b.s;
```

```
    }

    bool operator<(String b)
    {
        return s < b.s;
    }

    bool operator>(String b)
    {
        return s > b.s;
    }
};

int main()
{
    String a("apple"), b("banana");

    if (a == b)
        cout << "Equal";
    else if (a < b)
        cout << "First is smaller";
    else
        cout << "First is greater";

    return 0;
}
```

OUTPUT

```
Output

First is smaller
```